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FURNACE-X2: Pipelined CRC Offload and Serialized Packet Classification

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 29 October 2026 (2026-10-29 22:43 UTC+8 Beijing Time)

Abstract

FURNACE-X2 integrates preserved FURNACE-X1 CRC32C and CRC64/NVME offload with Ethernet/IPv4/IPv6 header extraction, five-field Toeplitz RSS and a programmable 64-rule ternary classifier. Its 64-bit and 512-bit streaming interfaces forward payload unchanged and independently deliver CRC and metadata records. Software-versus-hardware-RTL numerical comparisons passed for both widths and seeds 17, 43 and 89, covering 6,144 seeded packets plus directed tests. L1.5 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA met the 10.0 ns Arria 10 GX target with WNS_setup = +1.484 ns and hold slack = +0.043 ns. L3 physical Arria 10 FPGA HIL passed three consecutive deterministic bridge diagnostic runs; these diagnostics do not establish packet-level software-versus-physical-hardware numerical parity. [E174–E179, E259–E261]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The top module, furnace_x2_top, admits one outstanding packet. Accepted ingress concurrently feeds the unchanged CRC core and a 256-byte header capture. Registered byte scanning validates the packet and constructs its header and RSS tuple; bit-serial hashing and sequential rule comparison produce metadata. Serialization bounds logic depth rather than maximizing packet throughput. Configuration remains coherent until final payload, CRC and metadata consumption have all completed. All six retained X1 modules remain byte-identical to their reference sources. [E001–E012]

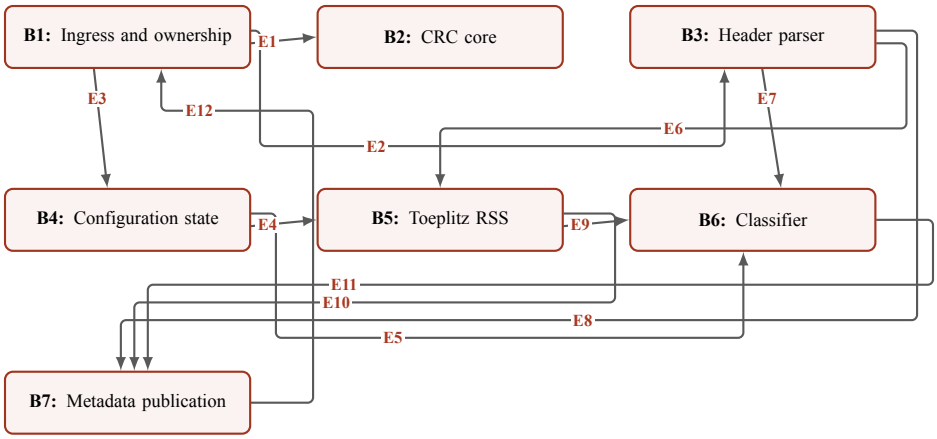


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Ingress and ownership	Jointly gates external ingress readiness and CRC input validity, prioritizes idle configuration and tracks three independent terminal handshakes.
B2: CRC core	Instance crc_core preserves furnace_x1_top, its pipeline hierarchy, payload forwarding and independent CRC consumption.
B3: Header parser	Instance parser captures 256 byte positions, counts accepted bytes and scans registered storage to validate headers.
B4: Configuration state	Maintains the RSS key, default action and 64 enabled value/mask/action entries; updates are idle-only.
B5: Toeplitz RSS	Instance rss processes one tuple bit per edge on a fixed 296-bit schedule.
B6: Classifier	Instance classifier performs a registered rule read, eight 64-bit comparisons and a priority commit for each entry.
B7: Metadata publication	Combines held header, length, hash and classification; retains the complete record until accepted.

Directed relationship	Meaning
E1: B1 to B2	Accepted payload and CRC controls
E2: B1 to B3	Identical accepted ingress beats
E3: B1 to B4	Idle configuration transfers
E4: B4 to B5	Coherent RSS key
E5: B4 to B6	Rules and default action
E6: B3 to B5	Tuple and bit count
E7: B3 to B6	Held classifier key
E8: B3 to B7	Header and packet length
E9: B5 to B6	Completion starts classification
E10: B5 to B7	Hash and eligibility
E11: B6 to B7	Hit, index and action
E12: B7 to B1	Metadata terminal handshake

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single rising-edge clock and active-low asynchronous reset. Every public output, including ready signals, is zero during reset.
s_axis_tdata, s_axis_tkeep, s_axis_tvalid, s_axis_tlast; s_axis_tready	Input; output ready	DATA_WIDTH, DATA_WIDTH/8, 1, 1; 1	Unsigned ingress stream; lane zero is the first transmitted byte. Transfers sample valid and ready before the rising edge.
s_check_enable, s_expected_crc32c, s_expected_crc64_nvme	Input	2, 32, 64 bits	First-accepted-beat checking controls. Enable bits select CRC32C and CRC64/NVME comparison.

Signal / group	Direction	Width	Contract
m_axis_tdata, m_axis_tkeep, m_axis_tvalid, m_axis_tlast; m_axis_tready	Output; input ready	DATA_WIDTH, DATA_WIDTH/8, 1, 1; 1	Unmodified, ordered payload and keep stream. Valid and payload remain stable while stalled.
m_crc_valid, m_crc32c, m_crc64_nvme, m_crc_mismatch, m_crc_protocol_error; m_crc_ready	Output; input ready	1, 32, 64, 2, 1; 1	One independently consumed CRC record per accepted final beat. Protocol error suppresses mismatch bits.
cfg_valid, cfg_op, cfg_index, cfg_value, cfg_mask, cfg_action, cfg_enable, cfg_rss_key; cfg_ready	Input; output ready	1, 2, 6, 512, 512, 32, 1, 328; 1	Atomic idle configuration. Operations 0–3 replace one rule, replace the RSS key, disable every rule and set the default action, respectively.
m_meta_valid, m_header, m_packet_bytes, m_rss_hash, m_rss_valid, m_rule_hit, m_rule_index, m_action; m_meta_ready	Output; input ready	1, 512, 32, 32, 1, 1, 6, 32; 1	One independently consumed metadata record per completed packet, including empty, malformed and unsupported traffic.

Parameter	Default	Meaning / supported values
DATA_WIDTH	512	Exactly 64 and 512 are supported and functionally exercised. Negative tests cover -1, 0, 8, 32, 128 and 1024: diagnostic-only elaboration reaches explicit PARAMETER ERROR with port widths of at least one bit.

III Behavior and corner cases

A Reset, configuration and ownership

Reset aborts pending traffic and results, disables all rules, restores the default RSS key and clears the default action. Idle cfg_valid takes priority over ingress. Configuration cannot change while a packet or result is outstanding. No next packet starts until final payload, CRC and metadata handshakes have all occurred; simultaneous terminal handshakes permit ingress on the following edge. Metadata computation continues despite blocked legacy consumers. Producers must hold stalled inputs stable. [E001, E002, E012]

B Header representation

m_header is the classifier key: source/destination IP addresses [127:0]/[255:128]; source/destination ports [271:256]/[287:272]; protocol [295:288]; version [299:296]; IP-valid/L4-valid/fragmented [300]/[301]/[302]; VLAN count [304:303]; outer/inner VID [316:305]/[328:317]; effective EtherType [344:329]; source/destination MAC [392:345]/[440:393]; parse_error/unsupported [441]/[442]. Bits [511:443] are zero. Multi-byte values use network numerical order; IPv4 addresses are zero-extended. [E002, E009]

C Supported parsing

Supports Ethernet II, zero through two 0x8100/0x88a8 VLAN tags, IPv4 IHL 5−15, TCP/UDP and IPv6 with up to eight Hop-by-Hop, Routing, Destination Options, Fragment or AH headers. Options/routing lengths are (HdrExtLen+1)<<3, Fragment length is eight bytes and AH length is (PayloadLen+2)<<2. Declared IP extents must fit received data; Ethernet padding is ignored. TCP requires offset at least five and its complete declared header. UDP requires length at least eight; its length may exceed a first fragment when more fragments follow. Non-initial fragments never expose ports. Atomic IPv6 fragments are not marked fragmented. No Next Header is a supported transport-free terminal. [E002, E009]

D Errors and bounded extraction

Packet bytes count asserted keep bits, saturating at 0xffffffff. Sizes outside 14−9216 bytes, partial non-final keeps and non-prefix final keeps set parse_error; zero final keep is legal. Wrong versions, reserved IPv4 fragment flags, repeated IPv6 Fragment headers and contradictory or truncated header lengths are errors. Error records clear extracted fields except parse_error and independently established unsupported status. Non-IP traffic, a third VLAN, unsupported transport, ESP, jumbograms, excessive extension chains and extraction beyond byte 255 produce unsupported records as applicable. Payload and CRC still complete; every record is classified. [E002, E009]

E CRC arithmetic

CRC32C uses reflected polynomial 0x82f63b78; CRC64/NVME uses 0x9a6c9329ac4bc9b5. Both use all-one initialization and final XOR. ASCII 123456789 yields 0xe3069283 and 0xae8b14860a799888; empty input yields zero. Malformed masks preserve X1 behavior: process full-width beats with invalid lanes zero-filled, then undo zero-byte transitions for missing final lanes, rather than compacting valid bytes. [E002, E004−E008, E014]

F Five-field RSS

Eligibility requires valid IP and TCP/UDP headers without error, unsupported status or fragmentation. The tuple is source address, destination address, source port, destination port and protocol: 13 bytes for IPv4, 37 for IPv6. Tuple and key bits are MSB-first; each asserted tuple bit selects a sliding 32-bit key window for XOR into an initially zero hash. There is no reflection, final XOR or modulo. Ineligible records return validity and hash zero. The 328-bit key accommodates the protocol byte, unlike conventional four-tuple RSS. Default key: 6d5a56da255b0ec24167253d43a38fb0d0ca2bcbae7b30b477cb2da38030f20c6a42b73bbeac01fa00. [E002, E010, E014]

G Ternary classification

An enabled entry matches iff ((header XOR value) AND mask)==0; mask one means care. Lowest index wins. Zero mask is wildcard; disabled entries cannot win. No match returns hit/index zero and the configured default action. Reserved header bits remain care-capable. Actions are opaque metadata, not payload operations. [E002, E003]

H Exact edge timing

Let E0 accept final ingress, with pre-edge handshakes and post-edge state labels. E1 registers byte zero; E2–E257 consume bytes 0–255; E258 publishes parsing. RSS processes bits 0–295 at E259–E554 and publishes at E555. Classifier rule r reads at E(556+10r), compares eight chunks and commits at E(565+10r). Rule 63 publishes at E1195; public metadata becomes valid post-E1196 and can transfer at E1197. At 100 MHz, validity occurs 11.96 microseconds after final ingress. Unstalled X1 payload/CRC appears after sixteen captures, post-E15 for acceptance at E0. [E001, E002]

IV Verification methods and measured results

A Software-versus-hardware-RTL numerical correctness

Method. Independent Python byte/field parsing, sliding-integer Toeplitz windows and bitwise polynomial CRC arithmetic compared against compiled Verilator results.

Result. Both widths passed seeds 17, 43 and 89: 1,024 seeded packets per width/seed, totaling 6,144 excluding directed traffic and reruns. Numerical comparisons concern simulated RTL, not physical packet vectors.

Evidence: E014, E175, E176, E177

B Protocol, priority and handshake coverage

Method. Directed packets, all rule winners, all 2,016 unordered matching-rule pairs, public-interface bit tests and independently stalled channels.

Result. Passed mandatory coverage, including final keep lengths, lane/capture/maximum-size boundaries, parser failure categories, all 512 classifier bits, RSS bit ordering, reset/configuration interactions and exact E1196 publication.

Evidence: E015, E017, E175, E176, E177

C Icarus behavior and parent compatibility

Method. Compact real Icarus/vvp runs at both widths with parent/copy comparison and independent numerical assertions.

Result. Passed startup/reset, multibeat traffic, independent output backpressure and exact metadata timing.

Evidence: E174

D Defect sensitivity

Method. Disposable RTL mutants compiled and executed against ordinary observable-result assertions.

Result. Ten mutants detected: CRC polynomial, parser endian, transport offset, fragment eligibility, RSS key direction, RSS tuple direction, mask polarity, reversed priority, dropped metadata and unstable metadata. Compilation errors were not counted as detection.

Evidence: E178

E Synthesizable substantive implementation

Method. Real Yosys synthesis at widths 64 and 512 with hierarchy, connectivity and mapped structural assertions.

Result. Passed zero-latch and zero-division/remainder checks, retained state and connected CRC/parser/RSS/classifier structure, including implemented 64-entry storage.

Evidence: E179

F Clean-directory reproduction

Method. Staged 23 declared source files and rebuilt Icarus and Verilator simulations using the same oracle.

Result. Passed five smoke packets per width, 128 seeded packets per width and 520/574 directed packets at widths 64/512. This bounded reproduction is separate from the full matrix.

Evidence: E257

G L1.5 RTL timing lint

Method. Native structural hazard analysis.

Result. Passed with zero divider/modulo, mixed-assignment, LUT clock-gate or AST loop-unroll hazards.

Evidence: E259

H L2 device timing

Method. Intel Quartus Prime 17.1 TimeQuest post-fit STA for the default-parameter top on Arria 10 GX 10AX115N2F45E1SG at 10.0 ns.

Result. Passed: WNS_setup = +1.484 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 117.43 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E260

I L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical hardware-bridge diagnostic runs.

Result. All three passed deterministically, recording diagnostic register 57 as 178 (0xB2). No packet-level physical CRC/header/RSS/classifier comparison vectors are supplied; software numerical parity is established separately against simulated RTL.

Evidence: E261, E175, E176, E177

J Material verification failures and supported resolution scope

Method. Surviving failure diagnostics considered separately from current oracle regressions and later isolated reproduction.

Result. The earlier behavior diagnostic records a metadata mismatch. Current oracle regression preserves independently discovered unsupported status alongside parse_error, and the current matrix passes; the absent original detailed run directory prevents independently establishing that mismatch's exact cause or repair. The earlier portability stderr records a malformed hexadecimal transcript containing an embedded DONE token, not an import failure. Its original detailed receipt directory is also absent. Later isolated reproduction passes, but does not establish the transcript failure's cause or a specific causal repair.

Evidence: E025, E014, E175, E176, E177, E153, E257

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 5	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

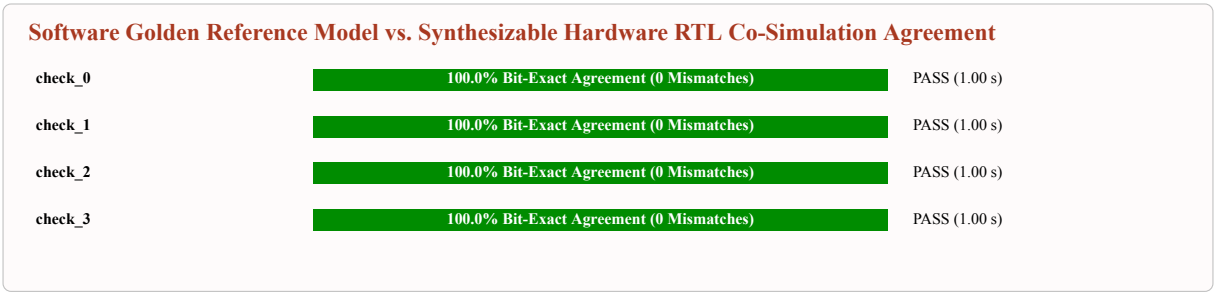


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +1.484 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_ across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

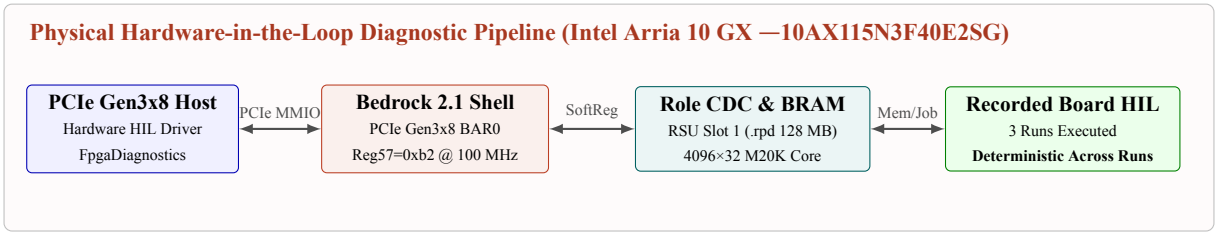


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile rtl/*.sv with furnace_x2_top selected as top and DATA_WIDTH set to 64 or 512. Use design/SPEC.md for the bit-level contract and design/ARCHITECTURE.md for module boundaries. Reference RTL supports compatibility verification and must not be compiled as duplicate product modules.

Assert reset before traffic and release safely relative to the clock. Configure through accepted cfg_valid/cfg_ready transfers. Supply checking controls with the first beat and keep stalled producer records stable. Handle payload, CRC and metadata readiness independently; any blocked terminal consumer prevents admission of the next packet.

Portable L1 dependencies are Python 3 standard library, Icarus/vvp, Verilator, a C++ compiler, make and Yosys. From the project root run python3 tests/verify.py. Partitions use --mode smoke, --mode behavior --seed 17 (also 43 and 89), --mode defects, --mode synthesis or --mode portability, with a project-relative --out root.

Verification invocations retain unique output children, command receipts, source hashes and assertion/coverage results. Native target STA and physical hardware gates are separate from portable L1; tests/verify.py requires no local cable or PCIe device identifier.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 10 Verilog/SystemVerilog files; 1286 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/furnace_classifier64.sv	124 total	122 code
rtl/furnace_crc_beat_pipeline.sv	94 total	91 code
rtl/furnace_crc_linear.sv	59 total	55 code
rtl/furnace_crc_output_stage.sv	94 total	90 code
rtl/furnace_crc_packet_accumulator.sv	96 total	92 code
rtl/furnace_crc_tail_pipeline.sv	80 total	77 code
rtl/furnace_header_parser.sv	412 total	409 code
rtl/furnace_rss_toeplitz.sv	59 total	59 code
rtl/furnace_x1_top.sv	124 total	121 code
rtl/furnace_x2_top.sv	171 total	170 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E003	rtl/furnace_classifier64.sv
E004	rtl/furnace_crc_beat_pipeline.sv
E009	rtl/furnace_header_parser.sv
E010	rtl/furnace_rss_toeplitz.sv
E012	rtl/furnace_x2_top.sv
E014	tests/oracle.py
E015	tests/scenarios.py
E017	tests/verify.py
E025	.vulcan/engineering/run-0001/attempt-0006/stdout.log
E153	.vulcan/engineering/run-0001/attempt-0101/stdout.log
E174	.vulcan/engineering/run-0001/attempt-0118/stdout.log
E175	.vulcan/engineering/run-0001/attempt-0119/stdout.log
E176	.vulcan/engineering/run-0001/attempt-0120/stdout.log
E177	.vulcan/engineering/run-0001/attempt-0121/stdout.log
E178	.vulcan/engineering/run-0001/attempt-0122/stdout.log
E179	.vulcan/engineering/run-0001/attempt-0123/stdout.log
E257	.vulcan/engineering/run-0001/attempt-0175/stdout.log
E258	.vulcan/engineering/run-0001/attempt-0176/response.json
E259	.vulcan/engineering/run-0001/rtl-timing-lint.json
E260	.vulcan/engineering/run-0001/quartus-sta.json
E261	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).